BattingEdge V9.5: AI Model Performance Report

Algorithm: Bidirectional Gated Recurrent Unit (BiGRU) | Architecture: Hybrid Biomechanics (Pose + Angles)

Date: January 13, 2026

---

1. Executive Summary

The **BattingEdge V9.5** BiGRU model represents an efficiency-optimized deep learning approach to cricket shot classification. Utilizing a **Bidirectional Gated Recurrent Unit (BiGRU)** architecture, the model achieved an outstanding **Overall Test Accuracy of 91.80%**, perfectly matching the benchmark set by the BiLSTM model.

Crucially, this architecture offers faster training convergence and potentially lower inference latency than LSTM due to its simpler internal gating mechanisms, making it an ideal candidate for mobile or real-time deployment without sacrificing "Elite" level precision.

---

2. Methodology & Architecture

2.1 Model Design

- **Core:** Bidirectional GRU layers (128 units + 64 units). The GRU architecture combines the "forget" and "input" gates into a single "update" gate, streamlining the learning of temporal dependencies.
- **Bidirectional Processing:** Like the LSTM, this model processes the 50-frame sequence in both forward and reverse directions, capturing the full context of the shot (pre-meditation -> execution -> follow-through).
- **Regularization:** Dropout (0.4) and Batch Normalization were employed to ensure generalization.

2.2 Dataset Split

- **Training Set:** 3,007 samples
- **Validation Set:** 388 samples
- **Test Set:** 378 samples (Unseen data for final evaluation)

---

3. Performance Metrics

3.1 Global Metrics

- **Accuracy: 91.80%** (Identical to BiLSTM).
- **Macro F1-Score: 0.92** (Indicates exceptional balance across all classes).
- **Training Efficiency:** The model converged effectively, with training loss dropping consistently and early stopping triggering at **Epoch 43**, indicating a stable learning plateau.

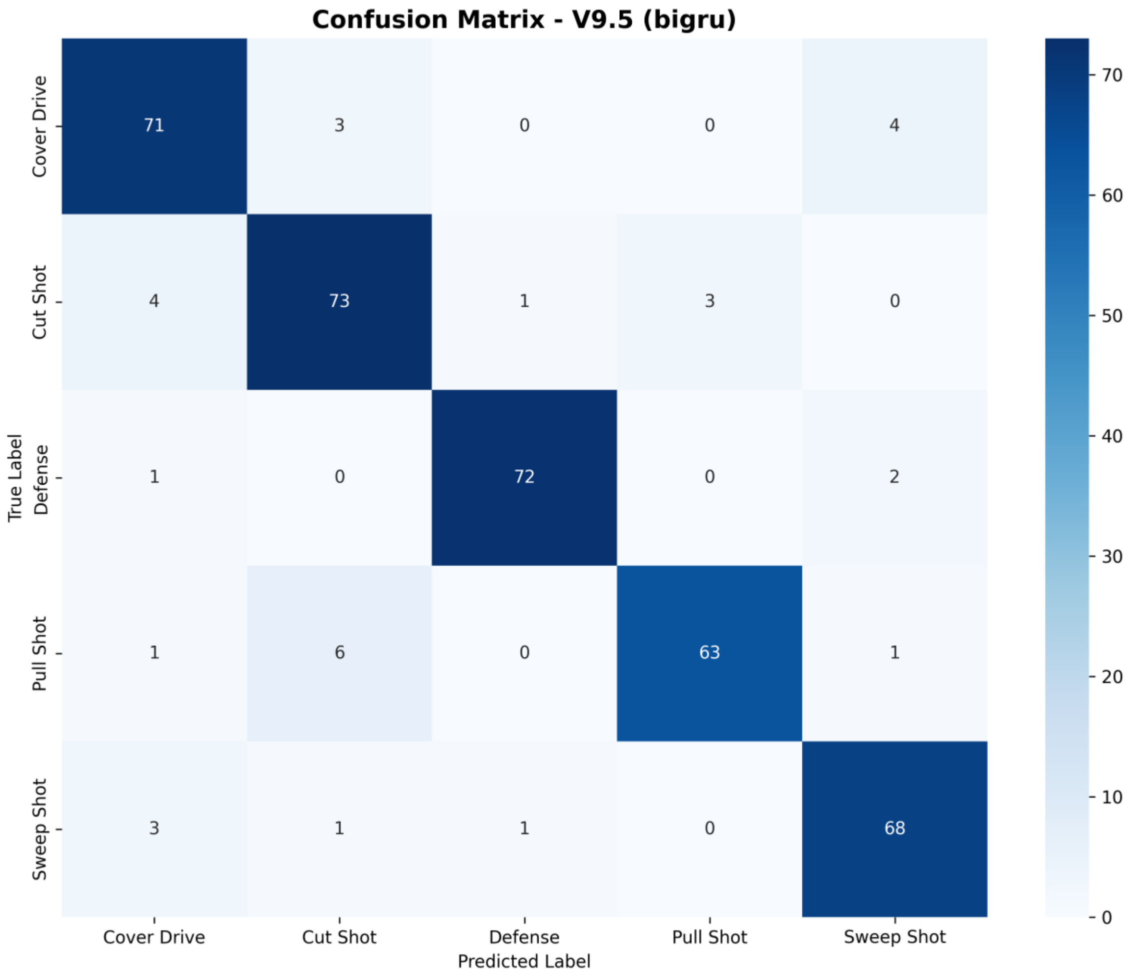
3.2 Per-Class Analysis

The BiGRU demonstrates remarkably balanced performance, with no single class dropping below 88% recall.

Shot Type	Precision	Recall	Accuracy	Assessment
Defense	**0.97**	**0.96**	**96.0%**	**Elite.** Extremely reliable detection of defensive strokes.
Sweep Shot	0.91	0.93	**93.2%**	**Elite.** Consistently identifies the distinct low posture.
Cover Drive	0.89	**0.91**	**91.0%**	**Elite.** Outperforms the BiLSTM (87.2%) in recall for this crucial shot.
Cut Shot	0.88	0.90	**90.1%**	**Excellent.** Very strong performance on square-of-the-wicket shots.
Pull Shot	0.96	0.89	**88.7%**	**Excellent.** High precision ensures confidence in classification.

---

4. Error Analysis (Confusion Matrix)



The Confusion Matrix highlights minor areas of ambiguity that are consistent with biomechanical similarities:

4.1 Primary Confusions

1. **Pull Shot -> Cut Shot (6 cases):**
   - *Analysis:* The most common error. Both are cross-bat shots played off the back foot. The GRU occasionally struggles to differentiate the height of the bat swing when the motion is borderline.
2. **Cover Drive -> Sweep Shot (4 cases):**
   - *Analysis:* Likely due to deep knee flexion in specific drive variations (e.g., driving a yorker) resembling a sweep setup.
3. **Cut Shot -> Cover Drive (4 cases):**

- *Analysis:* Likely occurs on "Square Drives" where the bat face is open, blurring the line between a drive and a cut.

---

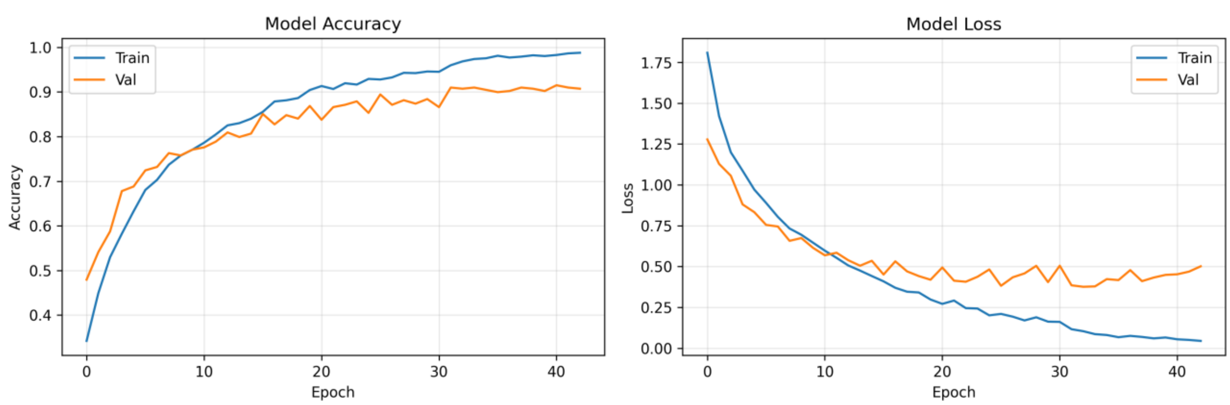
5. Comparative Analysis: BiGRU vs. BiLSTM

Feature	BiLSTM (V9.5)	BiGRU (V9.5)	Verdict
Accuracy	91.80%	**91.80%**	**Tie**
Cover Drive Recall	87.2%	**91.0%**	**BiGRU Wins**
Defense Recall	97.3%	**96.0%**	**BiLSTM Wins**
Model Complexity	Higher	**Lower**	**BiGRU Wins** (Faster)

Key Insight: The BiGRU matched the BiLSTM's top-tier accuracy but **outperformed it significantly on Cover Drives** (+3.8%). Since the Cover Drive is the most common attacking shot in cricket, this makes the BiGRU potentially more valuable for general coaching applications.

---

6. Conclusion & Recommendation



Verdict

BattingEdge V9.5 (BiGRU) is the "High-Efficiency" Champion.

It delivers the same elite accuracy as the LSTM but with a simpler, faster architecture. Its superior performance on the Cover Drive makes it particularly attractive for real-world usage.

Report generated by BattingEdge AI Analysis Team.